

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2022-23

Paper: BVCA-301 — Analog Digital Communication

Time: Three Hours

Maximum Marks: 70

Instructions: Attempt all questions. Write your answers in your own words as far as practicable.

SECTION — A

Long-answer questions to be answered in about 750 words each. Attempt any two: **15 × 2 = 30 Marks**

Question 1. Describe Analog and Digital Communication systems. Discuss the fundamentals of Digital Modulation.

OR

Describe 8-QAM and 16-QAM with suitable examples.

Question 2. Describe the detection of signals in AWGN.

OR

Describe Error Correcting Codes with reference to Convolutional Codes.

SECTION — B

Short-answer questions to be answered in about 200 words each. Attempt any six: **5 × 6 = 30 Marks**

- (a) Describe the classification of Signals.
- (b) Discuss PAM with examples.
- (c) Describe Linear Block Codes with examples.
- (d) What is Inter-Symbol Interference (ISI)?
- (e) Describe different wireless standards.
- (f) Discuss IEEE 802.11a and 802.11n.
- (g) Describe the classification of systems.
- (h) Discuss Hamming Codes.

SECTION — C

Answer all parts in a word or sentence each:

10 × 1 = 10 Marks

- (a) Full form of AWGN.
- (b) Define Parity.

- (c) Is the set of codewords $\{0000, 0101, 1010, 1111\}$ cyclic?
- (d) Define Network Communication.
- (e) What is the function of an Input Transducer?
- (f) Define the BIBO stability criterion.
- (g) Define Phase Modulation.
- (h) Define Pulse Time Modulation.
- (i) Define a Phasor Diagram.
- (j) Define a Constellation Diagram.